Eine Woche, ein Beispiel

4.13 lattices defining abelian variety

Ref:

[Deb99]: Complex tori and abelian varieties
[Mum74]: Mumford, David, Abelian varieties. Oxford university press Oxford, 1974.
[LR22]: Herbert Lange and Rubí E. Rodríguez. Decomposition of Jacobians by Prym Varieties. 2310.
[BL04]: Christina Birkenhake, and Herbert Lange. Complex Abelian Varieties. 2nd augmented ed. Grundlehren Math. Wiss. Berlin: Springer, 2004.

This document try to work out [Deb99, p28, Ex (3)]

Claim 1. [Mum74, p35, (1) $\Leftrightarrow$ (4)]
Let $\Lambda \subseteq \mathbb{C}^g$ be a full lattice. Then

$\mathbb{C}^g/\Lambda$ is an abelian variety
$\Leftrightarrow$ $\exists$ an $\mathbb{R}$-bilinear alternating form $\omega : \mathbb{C}^g \times \mathbb{C}^g \longrightarrow \mathbb{R}$ s.t.

$$\begin{cases} \omega(\Lambda \times \Lambda) \subseteq \mathbb{Z} \\ \omega(x, ix) > 0 \quad \forall x \neq 0 \\ \omega(ix, iy) = \omega(x, y) \end{cases} \qquad (*)$$

i.e. an integral Kähler form
$$\omega(u,v) = \operatorname{Re} h(iu,v) = \operatorname{Im} h(u,v) \qquad h(au,v) = \bar{a}\, h(u,v)$$
$$= -\operatorname{Re} h(u,iv)$$

From now on. suppose $\Lambda = \langle \lambda_1, \dots, \lambda_{2g} \rangle_{\mathbb{Z}}$, we denote

$$\Pi := (a_{ij})_{i,j=1}^{2g} := (\lambda_1^*, \dots, \lambda_{2g}^*)^T \qquad \Rightarrow \lambda_i^* = \sum a_{ij} e_j^*$$

The matrix Π encodes all information of the lattice (add a basis)

Q: For what kind of conditions of Π, can we find ω satisfying $(*)$?

Let $\omega = \sum_{i<j} c_{ij}\, \lambda_i^* \wedge \lambda_j^*$, then

$$\omega(\Lambda \times \Lambda) \subset \mathbb{Z} \iff c_{ij} \in \mathbb{Z} \qquad \forall i,j.$$

Write $x = \sum x_i e_i$, $y = \sum y_i e_i$, we get

$$\begin{aligned}
\omega(x,y) &= \sum_{k,l} x_k y_l\, \omega(e_k, e_l) \\
&= \sum_{k,l} x_k y_l \sum_{i<j} c_{ij}\ \lambda_i^* \wedge \lambda_j^* (e_k, e_l) \\
&= \sum_{k,l} x_k y_l \sum_{i<j} c_{ij}\ (a_{ik} a_{jl} - a_{jk} a_{il}) \\
\omega(x, iy) &= \sum_{k,l} x_k y_{l-g} \sum_{i<j} c_{ij}\ (a_{ik} a_{jl} - a_{jk} a_{il}) \\
&= \sum_{k,l} x_k y_l \sum_{i<j} c_{ij}\ (a_{ik} a_{j(l+g)} - a_{jk} a_{i(l+g)}) \\
\omega(ix, iy) &= \sum_{k,l} x_{k-g} y_{l-g} \sum_{i<j} c_{ij}\ (a_{ik} a_{jl} - a_{jk} a_{il}) \\
&= \sum_{k,l} x_k y_l \sum_{i<j} c_{ij}\ (a_{i(k+g)} a_{j(l+g)} - a_{j(k+g)} a_{i(l+g)})
\end{aligned}$$

Therefore,

$\omega(x, ix) > 0$ is an open condition on Π.

$$\omega(ix, iy) = \omega(x,y) \iff \sum_{i<j} c_{ij} \left(\begin{vmatrix} a_{ik} & a_{il} \\ a_{jk} & a_{jl} \end{vmatrix} - \begin{vmatrix} a_{i(k+g)} & a_{i(l+g)} \\ a_{j(k+g)} & a_{j(l+g)} \end{vmatrix} \right) = 0$$

$$\forall\ k, l \in \{1, \dots, 2g\}$$

Claim 2. $\mathbb{C}^g / \Lambda$ is an abelian variety

$\Lambda = \langle \lambda_1, \dots, \lambda_{2g} \rangle_{\mathbb{Z}}$

$\Pi = (a_{ij})_{i,j=1}^{2g} = (\lambda_1^*, \dots, \lambda_{2g}^*)^T$

$\iff \exists\ c_{ij} \in \mathbb{Z}$ for all $i,j \in \{1, \dots, 2g\}$ s.t.

① $\sum_{i<j} c_{ij} \left(\begin{vmatrix} a_{ik} & a_{il} \\ a_{jk} & a_{jl} \end{vmatrix} - \begin{vmatrix} a_{i(k+g)} & a_{i(l+g)} \\ a_{j(k+g)} & a_{j(l+g)} \end{vmatrix} \right) = 0 \quad \forall k, l \in \{1, \dots, 2g\}$

② $\left(\sum_{i<j} c_{ij}\ (a_{ik} a_{j(l+g)} - a_{jk} a_{i(l+g)}) \right)_{k,l=1}^{2g}$ is def positive.

Rmk. The equations in ① are not linear independent.
There are at most $\binom{g}{2} \cdot \frac{4 \cdot 2}{4} = g(g-1)$ equations.

Reason: When $l = k$ or $l = k+g$, we get 0.
e.p. when $g = 1$, we get no condition.

When $l \neq k$, & $l \neq k+g$, denote $\{k, l, k+g, l+g\} = \{k_1, k_2, k_3, k_4\}$,

$$a_{ik_1} a_{jk_2} - a_{ik_2} a_{jk_1} - a_{ik_3} a_{jk_4} + a_{ik_4} a_{jk_3}$$

$k_1 \leftrightarrow k_2$	$k_3 \leftrightarrow k_4$	neg
$k_1 \leftrightarrow k_3$	$k_2 \leftrightarrow k_4$	neg
$k_1 \leftrightarrow k_4$	$k_2 \leftrightarrow k_3$	same

Cor. Since

$$\mathcal{A}_g \subseteq \{\mathbb{C}^n/\Lambda \text{ a.v}\} \cong \{\Pi \in \mathbb{R}^{2g \times 2g} \mid \Pi \text{ satisfies } ①② \text{ for some } (c_{ij})\} / GL_g(\mathbb{C})$$

We know

$$\dim \mathcal{A}_g = \frac{1}{2}\left((2g)^2 - g(g-1)\right) - g^2 = \frac{1}{2} g(g+1).$$

Rmk. [BL04, 4.2]

We are actually reconstructing the Riemann relations.

Let $X = \mathbb{C}^g/\Lambda$, $\quad \Lambda = \langle \lambda_1, \dots, \lambda_{2g} \rangle$, $\quad \lambda_i \in \mathbb{C}^g$

$$\Pi = (\lambda_1, \dots, \lambda_{2g})_{g\times 2g} \in M^{g\times 2g}(\mathbb{C})$$

For $\omega: \mathbb{C}^g \times \mathbb{C}^g \longrightarrow \mathbb{R}$ an nondeg alternating form, denote

$$J_D := (\omega(\lambda_i, \lambda_j))_{2g\times 2g} \in M^{2g\times 2g}(\mathbb{Z}).$$

$\omega(\Lambda \times \Lambda) \subseteq \mathbb{Z}$

Then,

① $\omega(ix, iy) = \omega(x,y) \iff \Pi J_D^{-1} \Pi^T = 0$ (RC1)

② $\omega(x, ix) > 0 \quad \forall x \neq 0 \iff i\Pi J_D^{-1} \Pi^H > 0$ (RC2)

$\Pi^H = \overline{\Pi}^T$

With the symplectic basis, $\mathcal{L} = \mathcal{L}(H, \chi)$ is of type D the Riemann relations become:

① $\Pi_2 D^{-1} \Pi_1^T - \Pi_1 D^{-1} \Pi_2^T = 0$

② $i\Pi_2 D^{-1} \Pi_1^H - i\Pi_1 D^{-1} \Pi_2^H > 0$

For $(\Pi_1, \Pi_2) = (Z, D)$, the Riemann relations become:

① $Z^T = Z$

② $\operatorname{Im} Z > 0$

Rmk. The geometrical meaning of Π:

① parametrization of cplx structures on torus;

② a "change of basis", period;

③ comparison between Betti coh & de Rham coh.

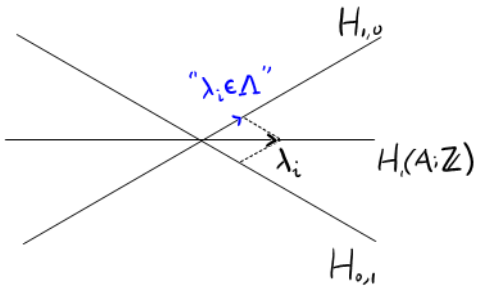
When we talk about "change of basis", it seems that they are not the basis of the same space.

$$(\lambda_1, \dots, \lambda_{2n}) = (e_1, \dots, e_n)\, \Pi$$

This is because we identify $H_1(A;\mathbb{R})$ with $H_{1,0}$ via projection. $\cong (H^{1,0})^*$

$$H_1(A;\mathbb{R}) \hookrightarrow H_1(A;\mathbb{C}) \cong H_{1,0} \oplus H_{0,1} \twoheadrightarrow H_{1,0}$$

And the actual change of basis is



$$(\lambda_1, \dots, \lambda_{2n}) = \text{“}2\mathrm{Re}\big((e_1, \dots, e_n)\,\Pi\big)\text{”}$$

$$= (e_1, \dots, e_n, \bar{e}_1, \dots, \bar{e}_n) \begin{pmatrix} \Pi \\ \cdots \\ \bar{\Pi} \end{pmatrix}$$

Write $\mathcal{T} = \begin{pmatrix} \Pi \\ \bar{\Pi} \end{pmatrix} \overset{\text{typical}}{=\!=} \begin{pmatrix} \tau & D \\ \bar{\tau} & D \end{pmatrix}$, we have

$$J_\Pi := \mathcal{T} J_D^{-1} \mathcal{T}^T = \begin{pmatrix} \Pi J_D^{-1} \Pi^T & \Pi J_D^{-1} \Pi^H \\ \bar{\Pi} J_D^{-1} \Pi^T & \bar{\Pi} J_D^{-1} \Pi^H \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -iM \\ iM^T & 0 \end{pmatrix} \qquad J_\Pi^T = -J_\Pi$$

where

$$M := i \Pi J_D^{-1} \Pi^H \qquad M > 0 \quad M^H = M$$

J_Π is roughly the matrix of ω under the basis $\{e_1, \dots, e_n\}$. Especially,

$$(\mathrm{RC1}) \iff H_{1,0} \text{ is isotropic subspace.}$$